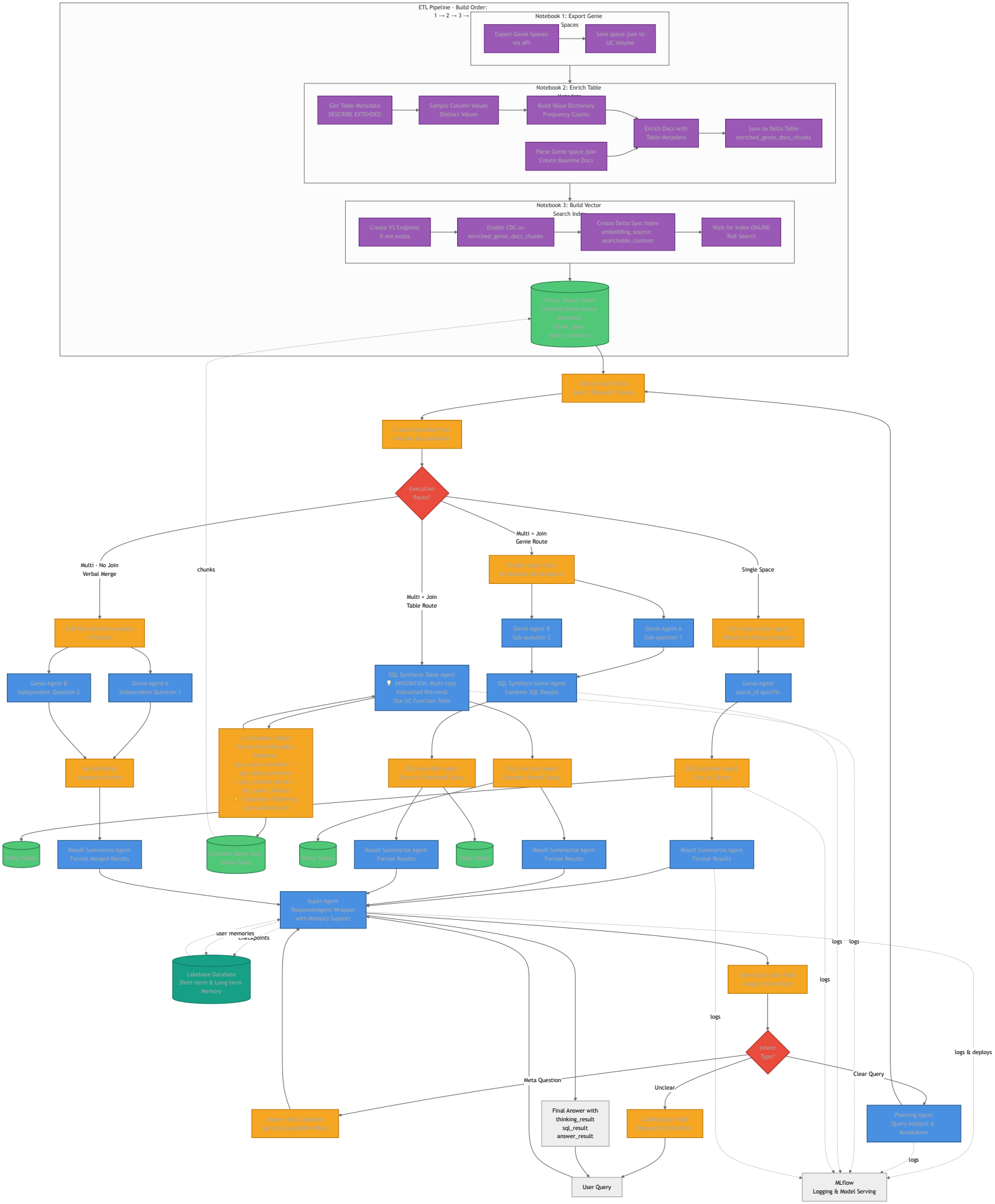
ETL Pipeline - Build Order:
1 → 2 → 3 →
Notebook 1: Export Genie Spaces
Export Genie Spaces via API
Save space.json to UC Volume
Notebook 2: Enrich Table Metadata
Get Table Metadata DESCRIBE EXTENDED
Sample Column Values Distinct Values
Build Value Dictionary Frequency Counts
Parse Genie space.json Create Baseline Docs
Enrich Docs with Table Metadata
Save to Delta Table enriched_genie_docs_chunks
Notebook 3: Build Vector Search Index
Create VS Endpoint If not exists
Enable CDC on enriched_genie_docs_chunks
Create Delta Sync Index embedding_source: searchable_content
Wait for Index ONLINE Test Search
Vector Search Index Enriched Genie Space Metadata chunk_type: space_summary
chunks
Vector Search Top Docs Relevant Spaces
Create Execution Plan Analyze Sub-questions
Execution Route?
Multi - No Join Verbal Merge
Multi + Join Table Route
Multi + Join Genie Route
Single Space
Parallel Route Calls to Multiple Genie Agents
Call Multi Genie Agents in Parallel
Genie Agent B Independent Question 2
Genie Agent A Independent Question 1
Genie Agent B Sub-question 2
Genie Agent A Sub-question 1
Call Single Genie Agent Databricks ResponsesAgent
Genie Agent space_id specific
SQL Synthesis Table Agent
INNOVATION: Multi-step Instructed Retrieval Use UC Function Tools
SQL Synthesis Genie Agent Combine SQL Results
Verbal Merge Prepare Answers
UC Function Toolkit Metadata Retrieval
get_table_overview
get_column_detail
get_space_details
Enhancing Following Query Generation
SQL Execution Agent Execute Combined Query
SQL Execution Agent Execute Joined Query
SQL Execution Agent Execute Query
Delta Tables
Result Summarize Agent Format Merged Results
Enriched Genie Doc Delta Table
Delta Tables
Result Summarize Agent Format Results
Delta Tables
Result Summarize Agent Format Results
Result Summarize Agent Format Results
Super Agent ResponsesAgent Wrapper with Memory Support
user memories
checkpoints
Lakebase Database Short-term & Long-term Memory
logs
logs
logs
Intent Detection Node Analyze Intent Type
logs
Intent Type?
logs & deploys
Clear Query
Meta Question
Unclear
Answer Meta Question System Capabilities Node
Final Answer with thinking_result sql_result answer_result
Clarification Node Request Clarification
Planning Agent Query Analysis & Breakdown
logs
User Query
MLflow Logging & Model Serving